

WEAVE: Escaping the Identity Shortcut in Lightweight Style Transfer with Wavelets

Anonymous ICME submission

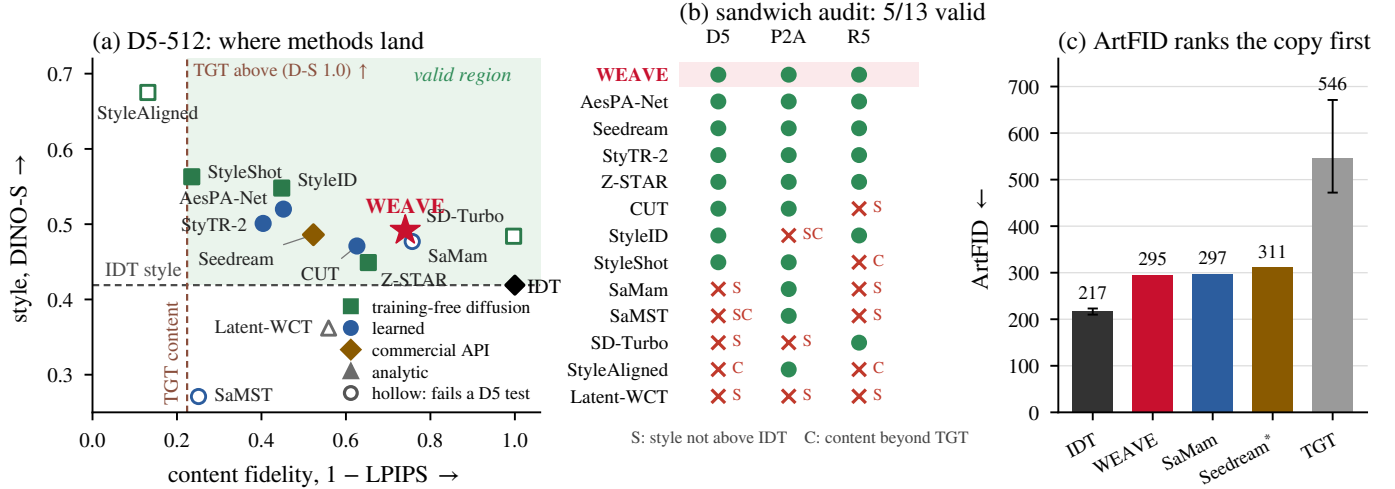


Fig. 1. The identity shortcut and the IDT-TGT sandwich. IDT returns the source; TGT returns a reference image of the target style. (a) On D5-512, valid transfer must lie above IDT in style and right of TGT in content; hollow markers fail at least one of the four tests (b), e.g., SD-Turbo and SaMam fall below IDT in CLIP-S. (b) Across three benchmarks only five of thirteen methods are valid everywhere. (c) ArtFID ranks the unchanged input first (*Seedream: 720 of 750 requests; intervals: bootstrap for IDT, 30 reference draws for TGT).

Abstract—Lightweight style transfer suits consumer devices, yet compact models often return images whose style is no closer to the target than the unchanged input, and content metrics reward this identity shortcut. We expose it with two trivial references, the source and a reference image of the target style, which bracket valid transfer; on three benchmarks, eight of thirteen compared methods leave this bracket. We trace the shortcut to the training signal: in latent velocity matching, coarse structure carries most of the gradient energy but is the least style-discriminative band. WEAVE resolves this in the Haar subbands of a frozen autoencoder latent. A source-anchored target provably shrinks the structural motion, a compact network learns band-weighted texture transport, and stepwise statistical alignment injects high-frequency style. With 1.04M trainable parameters and 1.4 minutes of training on one consumer GPU, WEAVE stays valid on all benchmarks and preserves the most content among valid methods.

Index Terms—style transfer, flow matching, wavelet decomposition, evaluation protocol, efficient generation

I. INTRODUCTION

Style transfer has moved from research demos into camera apps, creative tools, and on-device media pipelines, where memory and latency are tight. The strongest results come from diffusion editors and image foundation models that run a billion-parameter denoiser per image, some needing more than 60 GB of GPU memory [1], [2]. Compact models fit these budgets [3]–[5]. This paper asks whether their outputs actually change style, and how to build a compact model that does.

The identity shortcut. Copying the input already reaches a CLIP-S of 0.693 on our main benchmark, higher than two recent compact models (0.582 and 0.618). It is also ranked first by ArtFID [6] (Fig. 1c) and is optimal for LPIPS [7]. We therefore report two trivial outputs with every metric: IDT returns the source and TGT returns a reference image of the target style. A valid transfer must beat IDT on style and stay closer to the source than TGT on content. Under this IDT-TGT sandwich, eight of thirteen compared methods fail on at least one benchmark (Fig. 1b): some are no more stylized than a copy, others discard as much content as the reference itself.

Why compact models take it. Style transfer must keep structure while changing texture, yet a velocity field regressed on the whole latent scores both with one loss. In Haar coordinates this loss is dominated by the coarse LL band, which carries 69.5% of the gradient energy across 5,000 cross-style pairs while being the least style-discriminative band: its between-/within-style variance ratio is 0.12, against 0.56 for the finest band HH (Fig. 2). The style signal is a small part of the update, and networks that fit low frequencies first [8], [9] easily ignore it. Large editors escape by overwriting the image; compact models settle near the source. Both failures stem from the same entanglement, which extra width does not remove.

WEAVE disentangles the two goals by assigning them to subbands of the latent of a frozen SD1.5 VAE [10]. A source-anchored endpoint restricts the requested LL motion to a

channel-wise affine change of the source, which we prove removes at least 91% of its energy and cuts LL’s share from 69.5% to at most 17%. A 1.04M-parameter velocity network learns band-weighted transport whose texture heads are steered by pooled, coordinate-free codes of the reference, and an AdaIN update aligns high-frequency statistics after every Euler step. Training takes 82.8 s on one RTX 3060 and stops at an internal gradient transition without image evaluation. WEAVE is the only compared model under 10M trainable parameters that is valid on all three benchmarks; among the five valid methods, it preserves the most content on every benchmark with 11–34 $\times$ fewer trainable parameters than the other learned ones.

Contributions. (1) *Evaluation*: the IDT–TGT sandwich, a metric-agnostic check in raw units that exposes the identity shortcut and content collapse, with an audit of thirteen methods on three benchmarks. (2) *Analysis*: a subband measurement and a provable bound that link low-frequency gradient dominance to the design of the transport target. (3) *Method*: wavelet-decoupled velocity matching with orientation-specific texture conditioning, stepwise high-frequency alignment, and metric-free stopping. (4) *Evidence*: the best content preservation among valid methods at 1.04M parameters and 1.4 training minutes, with matched ablations, three seeds, reference-pool resampling, zero-shot transfer, and a training-free plug-in for a frozen SD1.5 editor.

II. RELATED WORK

Arbitrary and lightweight style transfer. Gatys et al. optimized Gram statistics [11], [12]; AdaIN and WCT made transfer feed-forward [13], [14]. SANet, StyTr², and AesPA-Net add attention [15]–[17], ArtFlow avoids content leak with reversible flows [18], and CUT learns unpaired translation by patch contrast [19]. MicroAST uses micro encoders and a micro decoder [3], and Mamba-based SaMST and SaMam reduce sequence cost [4], [5], [20]. Frequency cues appear in FDD, which mixes content and style spectra [21], in AesFA, which decomposes features by frequency [22], and in the wavelet-corrected WCT² [23]. These methods separate frequencies inside a feed-forward decoder; WEAVE uses Haar subbands as the coordinates of a transport objective with per-band targets, weights, and updates, derived from a measured imbalance.

Diffusion editors and fast translation. Latent diffusion [10] enables SDEdit, Prompt-to-Prompt, StyleID, StyleAligned, InST, Z-STAR, IP-Adapter, and StyleShot [24]–[31]. SD-Turbo distills sampling to one step [32], and LBM performs one-step latent bridge matching [33]. All of them run a large backbone per image; WEAVE keeps only the frozen VAE, and our plug-in study tests its Haar operator inside such an editor.

Flow matching and frequency-aware generation. Rectified flow and flow matching [34], [35] regress velocities along endpoint interpolations, and Schrödinger bridges [36]–[39] handle unpaired domains. FDIT constrains frequency bands separately for translation [40], FreqFlow reweights frequencies for class-conditional generation [41], and wavelet diffusion

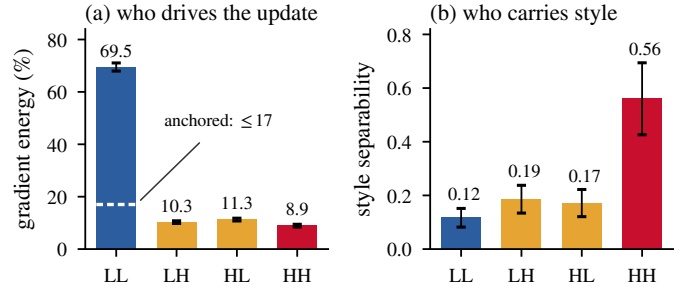


Fig. 2. Diagnosis on 5,000 cross-style D5 latent pairs. (a) Share of gradient energy per Haar band under the direct target; the dashed line marks the bound for the source-anchored target (Proposition 1). (b) Between-/within-style variance ratio of per-image band statistics. Error bars: standard error over the 20 style routes (a) and over styles (b).

accelerates sampling [42]. WEAVE changes the endpoint, basis, and band weights of the regression itself.

Evaluation. LPIPS [7] and ArtFID [6] are standard, and a recent review finds no consensus on style-transfer evaluation [43]. The sandwich adds two zero-cost references that give each metric an explicit direction.

III. DIAGNOSING THE IDENTITY SHORTCUT

A. The IDT–TGT Sandwich

A request consists of a source image x and a target style s , represented by a reference image r_s . We set $y_{\text{IDT}} = x$ and $y_{\text{TGT}} = r_s$ and call a method *valid* on a benchmark if its benchmark means satisfy

$$S(y) > S(y_{\text{IDT}}) \quad \forall S \in \mathcal{S}, \quad C(y) \succ C(y_{\text{TGT}}) \quad \forall C \in \mathcal{C}, \quad (1)$$

where $\mathcal{S} = \{\text{DINO-S}, \text{CLIP-S}\}$, $\mathcal{C} = \{\text{LPIPS}, \text{DINO-C}\}$, and $\succ$ means closer to the source. The first condition excludes the identity shortcut and the second excludes content collapse. Both references are free to compute, keep raw metric units, and apply to any metric pair; we check each inequality separately instead of forming a weighted score. They are empirical anchors, not bounds: a failure flags a regime for visual inspection on that benchmark. Copying is a strong baseline because content images already share features with any art collection: on D5, IDT scores DINO-S 0.419 and CLIP-S 0.693.

B. Spectral Dominance in Latent Velocity Matching

Let $\mathcal{W}(z) = (\ell, h_1, h_2, h_3)$ be the one-level orthonormal Haar transform of a latent $z \in \mathbb{R}^{4 \times H \times W}$, with LL band ℓ and the oriented bands LH, HL, and HH. Flow matching [34], [35] regresses a velocity v onto the displacement $u = z_1 - z_0$ between endpoints. By Parseval, $\|v - u\|^2 = \sum_b \|v_b - u_b\|^2$: the loss splits into independent band terms, and at initialization the share of band b in the gradient energy equals its share of the displacement energy, $\|u_b\|^2 / \|u\|^2$. Fig. 2 measures both on 5,000 cross-style D5 pairs with the direct target $z_1 = z_s$. LL holds 69.5% of the gradient energy, yet its style separability, the between-/within-style variance ratio of per-image band statistics, is the lowest (0.12 versus 0.56 for HH).

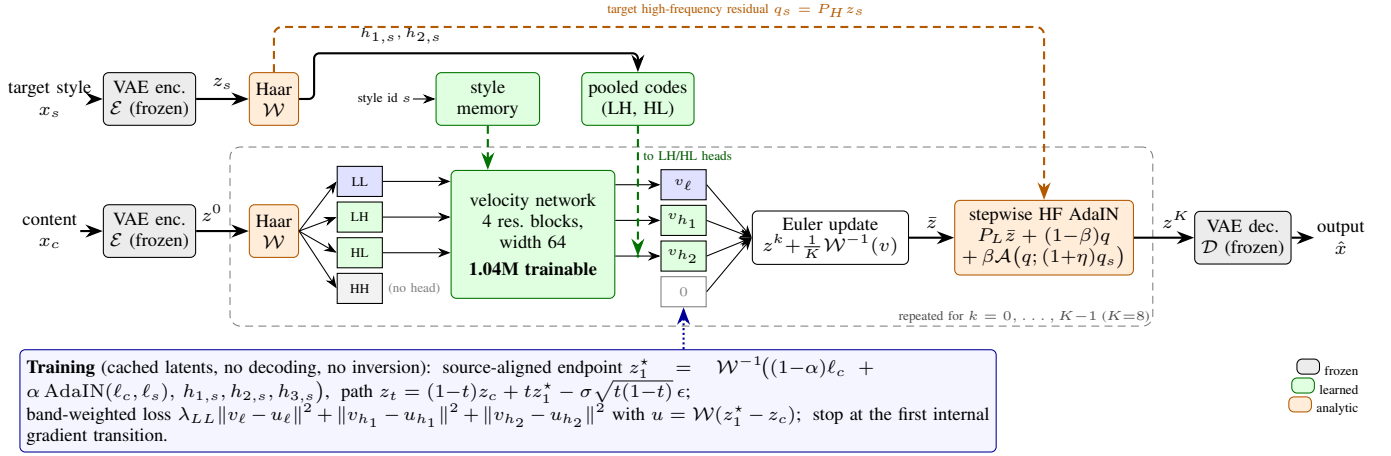


Fig. 3. WEAVE. A one-level Haar transform exposes LL and three high-frequency bands of the VAE latent. The compact network predicts LL/LH/HL velocities; pooled reference LH/HL codes modulate only the matching heads, and HH has no learned head. After every Euler update, AdaIN aligns the high-frequency residual with the reference statistics; LL is never altered by this step. Only the green blocks are trained.

The objective is driven by the band that matters least for style, and moving ℓ_c toward ℓ_s asks the network to reproduce the unrelated layout of x_s , which it can neither predict nor should copy.

IV. WEAVE

A. Setting

WEAVE is trained on unpaired image collections of K styles. A request names a target style by its index s and one reference image x_s ; at test time this is the first held-out image of the style, the same image used for TGT. We work on latents $z_c = \mathcal{E}(x_c)$ and $z_s = \mathcal{E}(x_s)$ of the frozen VAE, cached once before training, and write $P_L z = \mathcal{W}^{-1}(\ell, 0, 0, 0)$ and $P_H = I - P_L$ (Fig. 3).

B. Source-Anchored Endpoint

WEAVE anchors structure at the source and takes texture from the reference:

$$z_1^* = \mathcal{W}^{-1}((1-\alpha)\ell_c + \alpha \text{AdaIN}(\ell_c, \ell_s), h_{1,s}, h_{2,s}, h_{3,s}), \quad (2)$$

with $\alpha = 0.3$ and channel-wise spatial statistics in AdaIN [13].

Proposition 1: For one channel, let $\ell_c, \ell_s \in \mathbb{R}^N$ have means μ_c, μ_s , standard deviations σ_c, σ_s , and correlation ρ . The LL velocity requested by (2), $u_\ell = \alpha(\text{AdaIN}(\ell_c, \ell_s) - \ell_c)$, is an affine function of ℓ_c and

$$\|u_\ell\|^2 = \alpha^2 N[(\sigma_s - \sigma_c)^2 + (\mu_s - \mu_c)^2] \leq \alpha^2 \|\ell_s - \ell_c\|^2. \quad (3)$$

Proof: $\text{AdaIN}(\ell_c, \ell_s) - \ell_c = (\sigma_s/\sigma_c - 1)(\ell_c - \mu_c) + (\mu_s - \mu_c)$ gives the equality. Since $\|\ell_s - \ell_c\|^2 = N[\sigma_s^2 + \sigma_c^2 - 2\rho\sigma_s\sigma_c + (\mu_s - \mu_c)^2]$, the gap is $2N\sigma_s\sigma_c(1 - \rho) \geq 0$. ■

Two consequences follow. The requested structural motion only changes per-channel gain and offset, i.e., palette and contrast, and contains no layout of x_s . Because the high-frequency target is unchanged, averaging (3) over the pairs of Fig. 2 lowers LL's share of the displacement energy from 69.5% to at most 17.0%. Training samples $z_t = (1-t)z_c +$

$tz_1^* - \sigma\sqrt{t(1-t)}\epsilon$ with $t \sim \mathcal{U}[0, 1]$, $\epsilon \sim \mathcal{N}(0, I)$, and $\sigma = 0.02$; the label is the clean displacement $u = z_1^* - z_c$, and the perturbation, which vanishes at both endpoints, is absent at inference.

C. Band-Weighted Velocity Matching

The network $v_\theta(z_t, t, s, x_s)$ has a shared trunk of four width-64 residual blocks and separate 1×1 heads for LL, LH, and HL. With $(u_\ell, u_{h1}, u_{h2}, u_{h3}) = \mathcal{W}(u)$, it minimizes

$$\mathcal{L} = \mathbb{E}_t[\lambda_{LL}\|v_\ell - u_\ell\|^2 + \|v_{h1} - u_{h1}\|^2 + \|v_{h2} - u_{h2}\|^2], \quad (4)$$

with $\lambda_{LL} = 0.3$. The endpoint (2) changes the *requested* LL velocity, whereas λ_{LL} changes its optimization pressure. A style memory indexed by s gives the trunk a coarse prior. For image-specific texture, the reference LH and HL bands are pooled into coordinate-free, orientation-specific codes that modulate only the matching heads; feeding raw spatial reference bands instead leaks the reference layout (DINO-C falls to 0.40 in our probes; see supplement). HH has no learned head. The model has 1,037,087 trainable parameters.

D. Stepwise High-Frequency Alignment

Velocity matching organizes the motion but leaves a weak appearance signal. With $q = P_H z$, $q_s = P_H z_s$, and $\mathcal{A}(q; q_s) = \sigma(q_s)(q - \mu(q))/\sigma(q) + \mu(q_s)$ over channel-wise spatial statistics, each of the K inference steps is

$$\begin{aligned} \bar{z}^{k+1} &= z^k + \frac{1}{K} \mathcal{W}^{-1}(v_\ell^k, v_{h1}^k, v_{h2}^k, 0), \\ z^{k+1} &= P_L \bar{z}^{k+1} + (1 - \beta) \bar{q} + \beta \mathcal{A}(\bar{q}; (1 + \eta)q_s), \end{aligned} \quad (5)$$

where $\bar{q} = P_H \bar{z}^{k+1}$, $K = 8$, $\beta = 2$, and $\eta = 0.1$. With $\beta > 1$ the update over-relaxes toward the matched statistics; applied at every step, alignment becomes part of the trajectory rather than a post-process and supplies HH statistics without a learned head.

TABLE I

MAIN COMPARISON (D-S: DINO-S, C-S: CLIP-S, LP: LPIPS, D-C: DINO-C). EXCLUDING IDT AND TGT, THE BEST ENTRY PER COLUMN IS **BOLD** AND THE SECOND UNDERLINED. †: STYLE NOT ABOVE IDT; ‡: CONTENT BEYOND TGT; VALID: BENCHMARKS WITHOUT EITHER MARK. PARAMS IN MILLIONS, FROZEN PARAMETERS AS SUPERScript; PARAMS AND TRAIN RANK TRAINED METHODS. WEAVE USES ITS SELECTED D5 CHECKPOINT ON D5 AND DATASET-SPECIFIC CHECKPOINTS ON P2A/R5; COST REFERS TO THE D5 MODEL. TIMES ON ONE RTX 3060; *: AUTHOR REPORT ON FOUR GPUS [16]; ~3 h: Z-STAR ESTIMATED FROM STEP-SCALED RUNS (FULL 512 PX RUNS EXCEED 12 GB).

Method	D5-512				P2A-256				R5-WikiArt				Params	Train	Infer	Valid
	D-S†	C-S†	LP↓	D-C†	D-S†	C-S†	LP↓	D-C†	D-S†	C-S†	LP↓	D-C†	(M)	(min)	(750)	boards
IDT (identity)	0.419	0.693	0.000	1.000	0.416	0.663	0.000	1.000	0.433	0.731	0.000	1.000	0	free	0 s	–
TGT (target style)	1.000	0.863	0.776	0.215	1.000	0.835	0.800	0.169	1.000	0.796	0.747	0.230	0	free	0 s	–
SD-Turbo	0.484	0.693 [†]	0.003	0.922	0.341 [†]	0.674	0.603	0.279	0.505	0.767	0.449	0.538	0 ⁺¹¹⁰⁰	free	5 m	1/3
StyleAligned	0.675	0.780	0.869 [‡]	0.239	0.612	0.768	0.786	0.310	0.649	0.824	0.829 [‡]	0.315	0 ⁺¹¹⁰⁰	free	77 m	1/3
Z-STAR	0.449	0.784	0.347	0.549	0.498	0.786	0.332	0.552	0.514	<u>0.822</u>	0.384	0.526	0 ⁺¹¹⁰⁰	free	~3 h	3/3
StyleShot	<u>0.563</u>	<u>0.787</u>	0.765	0.377	<u>0.552</u>	<u>0.774</u>	0.713	0.335	0.484	0.787	0.795 [‡]	0.380	0 ⁺²⁴⁰⁰	free	5.1 h	2/3
StyleID	0.548	0.822	0.552	0.416	0.256 [†]	0.648 [†]	0.691	0.147 [‡]	<u>0.551</u>	0.815	0.493	0.385	0 ⁺¹¹⁰⁰	free	63 m	2/3
CUT	0.471	0.714	0.374	0.795	0.539	0.754	0.494	0.702	0.441	0.710 [†]	0.620	<u>0.795</u>	<u>7.0</u>	322.6	5 m	2/3
SaMST	0.271 [†]	0.618 [†]	0.749	0.145 [‡]	0.501	0.709	0.398	0.838	0.461	0.667 [†]	0.612	0.615	8.3	<u>39.5</u>	10 m	1/3
SaMam	0.477	0.582 [†]	<u>0.243</u>	<u>0.812</u>	0.505	0.677	0.205	0.870	0.503	0.712 [†]	0.227	0.925	8.8	436.0	17.6 m	1/3
StyTr-2	0.501	0.709	<u>0.596</u>	<u>0.747</u>	0.515	0.722	0.495	0.555	0.533	0.759	0.553	0.737	35.4 ^{+12.9}	~1440*	38 m	3/3
AesPA-Net	0.520	0.724	0.548	0.601	0.498	0.744	0.571	0.522	0.545	0.782	0.409	0.728	11.3 ^{+20.0}	–	23 m	3/3
Seedream 4.5	0.486	0.720	0.477	0.739	0.517	0.752	<u>0.227</u>	0.785	0.504	0.742	0.486	0.725	–	API	–	3/3
Latent-WCT	0.362 [†]	0.673 [†]	0.441	0.559	0.338 [†]	0.738	0.585	0.386	0.414 [†]	0.710 [†]	0.444	0.553	0	free	18 s	0/3
WEAVE (ours)	0.492	0.713	0.260	0.810	0.480	0.668	0.312	<u>0.861</u>	0.523	0.775	<u>0.289</u>	0.772	1.04 ⁺⁸⁴	1.4	<u>126 s</u>	3/3

E. Metric-Free Stopping

Training needs no decoding, inversion, or discriminator and runs 52 steps per epoch at batch 96. After each epoch, a probe batch of four fixed latents at $t = 0.5$ gives ρ_e , the ratio of LL-loss to HF-loss gradient norms on the shared trunk. Training stops at the first epoch in which the gate of the texture route grows and $\rho_e/\rho_{e-1} \leq 0.65$, i.e., when high-frequency learning takes over the trunk. On D5 the rule fires at epoch 4 (ρ drops from 1.09 to 0.16) after 82.8 s, without any image metric.

V. EXPERIMENTS

A. Setup

Benchmarks. *D5-512*: five WikiArt styles (Early Renaissance, Impressionism, Minimalism, Rococo, Ukiyo-e) at 512 px with 3,600 training images each; 30 held-out sources per style are sent to all five styles (750 requests, same-style requests kept). *P2A-256*: the same layout over photo, Monet, Van Gogh, Cézanne, and Hayao at 256 px. *R5-WikiArt*: five randomly drawn WikiArt families (Cubism, Expressionism, Pop Art, Romanticism, Symbolism) at 512 px.

Metrics. DINO-S (primary) is the maximum cosine similarity between the DINOv2-small CLS feature [44] of the output and those of up to 30 held-out references of the target style; it rewards resembling some member of the family rather than its average. CLIP-S is the CLIP ViT-B/32 [45] similarity to the target-style prototype, DINO-C the DINOv2 similarity to the source, and LPIPS uses AlexNet.

Baselines and implementation. We compare training-free diffusion editors (SD-Turbo [32], StyleAligned, Z-STAR, StyleShot, StyleID), learned models (CUT, SaMST, SaMam, StyTr², AesPA-Net), the Seedream 4.5 API [46], and an analytic Latent-WCT, each at its released or recommended

operating point. WEAVE uses AdamW (learning rate 2×10^{-4} , cosine decay), bf16, and one RTX 3060 (12 GB). Training time excludes one-off latent caching; inference time includes eight ODE steps and VAE decoding.

B. Main Results

The audit. Only five of thirteen methods are valid on all three benchmarks (Table I, Fig. 1b), and the failures are instructive. SD-Turbo attains the best D5 LPIPS (0.003) because it barely edits the input: its CLIP-S equals that of IDT. SaMam, the strongest compact baseline on content, has CLIP-S below IDT on two benchmarks, so its low LPIPS partly reflects the identity shortcut. StyleAligned has the highest DINO-S everywhere but moves farther from the source than the reference itself on D5 and R5. Latent-WCT never clears IDT: wavelet statistics alone do not transfer style, and the learned transport is necessary.

Valid methods. Among the five valid methods (Z-STAR, StyTr², AesPA-Net, Seedream 4.5, and WEAVE), WEAVE has the highest DINO-C on all three benchmarks (0.810/0.861/0.772) and the lowest LPIPS on D5 and R5. Its DINO-S lies within the range of the others on D5 and R5 and is 0.018 lower on P2A, where each Haar subband of the 256 px latent holds only 16×16 coefficients per channel. On D5, WEAVE matches the commercial Seedream 4.5 in DINO-S (0.492 vs. 0.486) at about half its LPIPS (0.260 vs. 0.477). Among all thirteen methods, no competitor beats WEAVE on a style and a content metric at once on D5 and R5, i.e., WEAVE is Pareto-optimal in all four style–content metric pairs there. On P2A, SaMam is better on all four metrics, albeit with $8.5 \times$ the parameters and over $300 \times$ the training time, and Seedream is better on style and LPIPS.

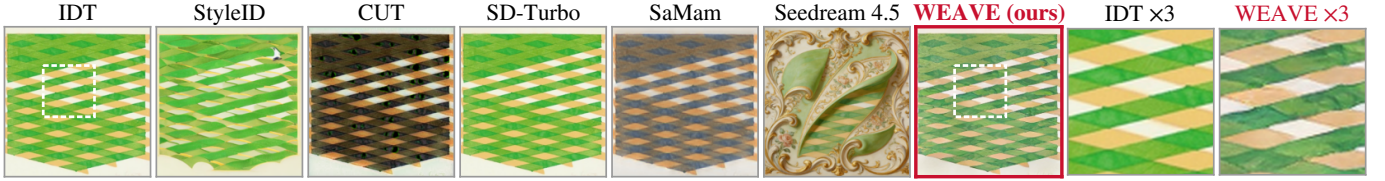


Fig. 4. Extreme request (Minimalism $\rightarrow$ Rococo). SD-Turbo returns the input (identity shortcut) and Seedream replaces the lattice with ornaments (content collapse). WEAVE keeps the lattice; the close-ups (dashed boxes, $3\times$) show the flat Minimalist fill turned into varied, painterly strokes in a warmer palette.

TABLE II
COMPONENT ABLATION ON D5-512 (750 REQUESTS).

Variant	DINO-S $\uparrow$	CLIP-S $\uparrow$	LPIPS $\downarrow$	DINO-C $\uparrow$
WEAVE (epoch 4)	0.4918	0.7128	0.2595	0.8102
w/o stepwise AdaIN ($\beta=0$)	0.4821	0.7005	0.2258	0.8455
w/o texture conditioning	0.4825	0.7157	0.2837	0.8125
direct endpoint ($z_1^*=z_s$)	0.4894	0.7172	0.3138	0.7966
$\lambda_{LL}=1.0$	0.4910	0.7180	0.2626	0.7874
learned HH head	0.4930	0.7164	0.2670	0.8061

Cost. WEAVE has the fewest trainable parameters and the shortest training among the learned methods we compare (1.4 min vs. 39.5–1440 min) and needs 168 ms per 512 px image including VAE decoding. Among valid methods, the next cheapest learned model (AesPA-Net) has $11\times$ more trainable parameters and $11\times$ longer inference.

Qualitative comparison. Fig. 4 shows both failure modes on an extreme request. The close-ups show what the metrics measure: WEAVE keeps the geometry of the source and changes its texture and palette.

C. Ablations

Table II changes one component at a time on the same 750 requests. Inference-time controls reuse the selected checkpoint; retrained variants (seed 42) are reported at their best-DINO-S epoch, which favors them. As a yardstick, DINO-S varies by 0.003 (s.d.) across seeds. Removing stepwise AdaIN moves the output toward the source (DINO-S -0.010 , CLIP-S -0.012 , LPIPS -0.034). Removing the texture conditioning lowers DINO-S by 0.009 and raises LPIPS by 0.024 while CLIP-S is nearly unchanged. Replacing the anchored endpoint by the direct target has the largest content cost (LPIPS $+0.054$, DINO-C -0.014), as Proposition 1 predicts, and its style peak arrives nine epochs later. Raising λ_{LL} to 1.0 changes peak DINO-S by less than one seed s.d. but lowers DINO-C by 0.023, and a learned HH head gives a within-noise style gain at a small content cost, so we keep the simpler model.

D. Robustness and Transfer

Seeds and stopping. The stopping rule selects epochs 4, 4, and 3 for seeds 42, 7, and 123 (DINO-S 0.490 ± 0.003 , LPIPS 0.261 ± 0.006); for the two seeds with full 15-epoch curves, the selected epoch is the DINO-S peak.

Reference pools. WEAVE conditions on one reference per style that also belongs to the DINO-S pool. We therefore redraw the pool: with 8 of the 30 references per style, sampled

TABLE III
BEYOND THE TRAINING SETTING. TOP: ZERO-SHOT ON FIVE UNSEEN WIKIART FAMILIES (750 PAIRS, D5 CHECKPOINTS). BOTTOM: HAAR ALIGNMENT IN A FROZEN SD1.5 EDITOR (600 PAIRS).

Method	DINO-S $\uparrow$	CLIP-S $\uparrow$	LPIPS $\downarrow$	DINO-C $\uparrow$
SaMST	0.3913	0.7485	0.6213	0.2680
SaMam	0.5466	0.7577	0.3990	0.7369
WEAVE	0.5313	0.7742	0.2802	0.7587
SD1.5 img2img + Haar alignment	0.3974	0.7221	0.3591	0.6589
	0.4080	0.7366	0.3630	0.6531

1,000 times so that the conditioning image is usually absent, the mean DINO-S margin over IDT stays positive in every draw (0.030, 95% interval $[0.020, 0.039]$) and close to the full-pool margin (0.033); about 69% of individual requests improve.

Unseen families. Applied without retraining to five held-out WikiArt families (Table III, top), the D5 checkpoint gives the best CLIP-S, LPIPS, and DINO-C; SaMam is 0.015 higher in DINO-S at a much higher LPIPS.

Plug-in for a frozen editor. Adding the same Haar-domain alignment (LH/HL/HH at scale 0.8, LL untouched) to a frozen SD1.5 image-to-image pipeline (empty prompt, strength 0.5, 20 DDIM steps) raises DINO-S and CLIP-S on 600 cross-style D5 pairs (paired t -test, $p < 10^{-15}$) at a small LPIPS change (Table III, bottom), for 4.9 ms per image and no weight update.

VI. CONCLUSION

Limitations. WEAVE uses one SD1.5 VAE and a single Haar level; styles defined by global layout or spanning several scales may need deeper decompositions, and at 256 px the subbands are coarse. Haar separates latent coordinates, not semantic factors, and the nonlinear decoder can couple them. The stopping rule is validated on one architecture and three seeds. IDT and TGT are benchmark-specific anchors measured with imperfect perceptual proxies, and baselines run at their recommended operating points rather than under a matched compute budget.

Conclusion. The identity shortcut hides behind metrics that reward it. Two trivial references make it visible, and under them most compared methods fail somewhere. Its cause is spectral, since coarse structure dominates the gradient but carries little style, and WEAVE removes it by anchoring structure at the source, learning texture transport in explicit subbands, and injecting high-frequency statistics along the

trajectory. The result is the most content-preserving valid transfer we measured, from a 1.04M-parameter model trained in 1.4 minutes, which makes faithful style transfer practical on consumer hardware. Code, checkpoints, and the 750-request protocol will be released.

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